Detecting Cannabis–Alcohol Substitution in Aggregate Sales Data: Statistical Power and the Limits of Small-Sample Synthetic Control

By Alex Ronczewski*

Nulls are the most common outcome of studies that look at whether the legalization of cannabis substitutes for state-level alcohol sales. This paper asks if such designs could detect substitution of plausible size. I use per-capita ethanol sales and the staggered opening of cannabis retail across the United States and estimate the pooled effect with six estimators; all six return positive-signed nulls between $+0.7$ *and* $+2.8$ *percent. Backdated placebos produce negative pseudo-effects of one to two percent when no state had legalized. I run a design-based power simulation which has two main results. First, the software-default inference for the primary estimator, a wild bootstrap, has a size of exactly zero in the simulation and rejects nothing at any effect up to twelve percent, so nulls produced under it carry little weight. The failure is mechanical: the estimated effect enters the bootstrap's replicate variance, so the t-statistic is bounded near one and cannot reach any critical value. A jackknife on the same draws works but is conservative, with a minimum detectable effect of roughly 7.2 percent. Second, the calibrated minimum detectable effect is* 4.5 *to* 6.4 *percent. Substitution effects below two to three percent are not detectable by these estimators, and a four-percent effect is significant in only the most sensitive one, leaving substitution in the low single digits mostly below reach, even when such effects are still relevant for policy decisions.*
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A common case for cannabis legalization is the claim that it substitutes for more harmful substances, with alcohol being the most common. There have been numerous studies on the large social cost of alcohol (Sacks et al., 2015; Cook, 2007), so even a small substitution effect from legal cannabis availability would

* The data is from the National Institute on Alcohol Abuse and Alcoholism Surveillance Report #122, state cannabis tax and regulatory portals (Colorado Department of Revenue, Washington State Liquor and Cannabis Board, Oregon Liquor and Cannabis Commission, Nevada Cannabis Compliance Board, California Department of Tax and Fee Administration, Massachusetts Cannabis Control Commission), the Bureau of Economic Analysis, the Bureau of Labor Statistics, the Federation of Tax Administrators, and the U.S. Census Bureau. Replication code and a one-command run script are available in the project repository. Claude (Anthropic) was used for code development and debugging; all research design decisions, data choices, and final text are my own. All errors are mine.

be a meaningful benefit. If there is no substitution effect, a common argument for legalization would be ruled out. In 2014 Colorado and Washington opened the first recreational cannabis retail markets and since then there has been a staggered roll-out across states. This has created substantial empirical literature on the cannabis–alcohol relationship, but that literature has not reached a consensus.

Anderson, Hansen and Rees (2013) find evidence consistent with a substitution effect in traffic fatality data. Baggio, Chong and Kwon (2020) estimate a roughly 12 percent decline in alcohol sales using retail-scanner data and a border-county design. Veligati et al. (2020) find no effect in aggregate state tax-receipt data, while Calvert and Erickson (2021) find mixed, beverage- and state-specific results in household scanner-panel data, which does not show a substitution effect or complementary relationship. Looking at the broader evidence shows that cannabis and alcohol act as both substitutes and complements depending on the population (Subbaraman, 2016; Risso et al., 2020). Studies that find substitution effects use granular data (scanner panels, individual fatality records); the ones built on state-level aggregates mostly find nulls.

This paper examines if these aggregate nulls show the absence of a true substitution effect or the limits of the data. The primary concern is statistical power: with seven legalized states, thirty never-treated control states and twenty annual observations per state, it is not certain if the estimator can distinguish a substitution effect from noise. If it lacks statistical power, then a null from this design carries little information, and reporting it as evidence against substitution overstates what the research design can claim.

This paper accomplishes this in three steps. First, I estimate the pooled effect of recreational retail opening on log per-capita ethanol sales with six estimators: partially-pooled augmented synthetic control (Ben-Michael, Feller and Rothstein, 2022), synthetic difference-in-differences (Arkhangelsky et al., 2021), generalized synthetic control (Xu, 2017), matrix completion (Athey et al., 2021), the group-time estimator of Callaway and Sant'Anna (2021), and two-way fixed effects. I use these six estimators to rule out the possibility that a single method's bias creates the aggregate nulls. Second, I backdate treatment to all possible placebo adoption years and then re-estimate; this measures the pseudo-effects each method produces in periods when no state has legalized. Third, I run a design-based power simulation: I create fake substitution effects and insert them into randomly drawn never-legalized states using the real staggered adoption dates, and record how often each procedure rejects the null.

The three steps are consistent in their findings. The six estimators return pooled estimates between $+0.7$ and $+2.8$ percent, positive and not significant. The backdated placebos find that every estimator produces a spurious effect of around -1 to -2 percent in a non-treatment window, so a true effect of that size is within the research design's noise. The power simulation shows that the inference conventionally reported for the primary estimator, the software's default wild bootstrap, rejects in zero percent of draws at every fake-inserted effect up to

12 percent, while a jackknife on the same draws works but is mildly conservative. With a correctly calibrated randomization test, the primary estimator's minimum detectable effect at 80 percent power is 6.4 percent of per-capita ethanol, which corresponds to roughly 3.2 standard drinks per adult per month. If we extend this to all our estimators, the smallest minimum detectable effect is 4.5 percent. The academic literature finds effects from null to twelve percent (Baggio, Chong and Kwon (2020) estimates twelve percent from scanner data); effects that are in the low single digits fall below the detection threshold of every estimator, even when effects of this size are relevant for policy discussion.

This paper has two main contributions. First, a statement of what one standard aggregate design can establish. The null found here is a statement about detectability rather than evidence that substitution isn't present, and it does not contradict the granular-data studies that find substitution. Other aggregate studies differ in data, treatment definitions, estimators, sample periods, and donor pools, so this paper's simulation does not settle their power; it shows how easily each of those designs could settle the question for itself. The second is broader and reaches past cannabis–alcohol. A design-based power audit is cheap to produce and, for small-sample staggered synthetic control, necessary: the estimator in this design is nearly unbiased, so its point estimates give no sign when the attached inference cannot reject. The power simulation here shows exactly that failure for the inference this estimator carries by default, a wild bootstrap whose standard errors are several times the estimator's true sampling dispersion and grow with the estimated effect itself (Section V.C), making nulls close to guaranteed regardless of the true substitution effect. This configuration is common in applied work: a handful of treated states matched to many donors over around twenty years, and a design-based power simulation makes the problem visible.

The difficulty of inference with few treated units is documented. Conley and Taber (2011) show that standard asymptotics fail when the number of policy changes is small and suggest building the null distribution from the untreated units, which is how the calibrated randomization tests in this paper are created. Ferman and Pinto (2019) characterize when few-treated-group inference over- or under-rejects under heteroskedasticity, and MacKinnon and Webb (2017, 2018) show that wild cluster bootstrap tests can under-reject aggressively when only a few clusters are treated. The zero-power result in Section V is therefore not a claim that small treated samples break standard inference; which has already been established in the literature. The contribution relative to this literature is twofold. First, the failure shown here is different: the default bootstrap in this implementation does not only under-reject: its t-statistic is bounded near one at every effect size, so its power is zero against every alternative by construction, and Section V.C traces the mechanism to the resampling scheme. Second, the diagnostic is different: a design-based power audit built from the design's own donor pool detects this type of failure directly, without test-specific asymptotic analysis, and prices the design in effect-size units at the same time.

The layout of the paper is as follows: Section I describes the outcome data, the treatment, and the first stage. Section II sets out the sample rules, the six estimators, and the inference. Section III presents the pooled and state-level estimates. Section IV reports the in-time placebo evidence. Section V presents the power analysis. Section VI is for robustness checks. Section VII discusses implications and our findings, and Section VIII is the conclusion.

I. Data and Treatment

A. Outcome: Per-Capita Ethanol Sales

The outcome variable is per-capita ethanol consumption, measured in gallons of pure ethanol per person aged 21 and older, from the National Institute on Alcohol Abuse and Alcoholism Surveillance Report #122 (Slater and Alpert, 2025). I construct the series from state alcohol tax receipts and shipment data, split into beer, wine and spirits. It runs to 2023. I take the natural logarithm so that the estimates read as percentage effects, and restrict the primary analysis window to 2000–2019; this excludes the COVID-19 period which shifted both the level and the composition of alcohol sales in ways unrelated to cannabis policy, and also gives each state at least fourteen pre-treatment years for matching. The full 2000–2023 window appears as an extension in Section VI. It is worth noting that the time series measures sales and not consumption. Alcohol purchased in one state and consumed in another is attributed to the state in which it was bought. New Hampshire, whose state-run liquor stores and absence of a sales tax draw heavy cross-border purchasing, is the best-known case, and the spillover robustness checks in Section VI address this problem.

Table 1 reports pre-treatment summary statistics. The treated states have baseline drinking levels between 2.6 gallons per adult (California, Washington) and 3.9 (Nevada), which shows that no single untreated state can serve as a credible comparison. I compute the donor average over 2000–2013 so the comparison rows cover the same years used for matching.

Figure 1 plots log per-capita ethanol for the early legalizers (Colorado and Washington), the later legalizers, and the never-legalized states. The early legalizers sit below the late legalizers, and the gap widens over time. The drift rules out a simple before-and-after comparison and motivates constructed counterfactuals; it is also, as Section IV shows, the design's central vulnerability.

B. Treatment Timing and the First Stage

The treatment is the opening of legal recreational retail as opposed to the ballot measure, as the retail opening is the moment consumer behavior responds and substitution effects can appear. The seven candidate treated states and their retail years are Colorado (2014), Washington (2014), Oregon (2015), Alaska (2016), Nevada (2017), California (2018), and Massachusetts (2018).

Table 1—Per-Capita Ethanol by Treated State, Pre-Treatment

State	Period	Mean	SD	Min	Max
Colorado	2000–2013	3.086	0.065	2.975	3.183
Washington	2000–2013	2.574	0.083	2.455	2.704
Oregon	2000–2014	2.833	0.146	2.598	3.012
Alaska	2000–2015	3.201	0.179	2.853	3.514
Nevada	2000–2016	3.902	0.163	3.628	4.137
California	2000–2017	2.600	0.052	2.487	2.684
Massachusetts	2000–2017	2.863	0.045	2.756	2.950
Donor average	2000–2013	2.613	0.631	1.509	5.394

Notes: Gallons of pure ethanol per person aged 21 and older, NIAAA Surveillance Report #122. Each treated state's row covers its own matching period, 2000 through the year before retail opening. The donor row covers 2000–2013, the matching window, across the 30 states that had no recreational retail through 2023.

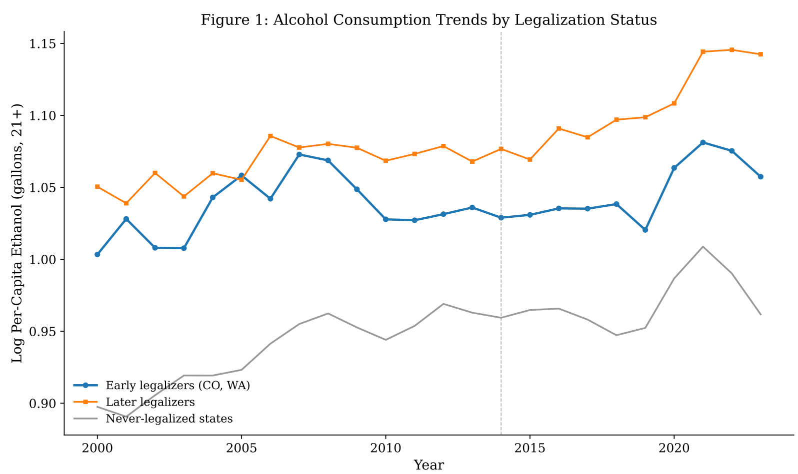

Figure 1. Alcohol Consumption Trends by Legalization Status

Notes: Mean log per-capita ethanol by group. Early legalizers are Colorado and Washington; later legalizers are Oregon, Alaska, Nevada, California, and Massachusetts; never-legalized states are the 30 states without recreational retail through 2023. The dashed line marks 2014.

Null results are only informative if the treatment actually changed access for citizens to recreational cannabis. I document the first stage directly with annual legal cannabis sales collected from each state's tax or regulatory authority.[1] Figure 2 shows per-adult legal sales. Retail opening produced large and fast access changes: Colorado reached roughly $213 per adult per year in its first two retail years and Nevada roughly $236. California is the exception. Its dose at opening was about $83 per adult, 2.6 times smaller than Colorado's, because California had operated a near-universal de facto medical access regime since 1996, so the 2018 retail opening changed the legal form of access more than its effective availability. This asymmetry matters for interpreting California's estimate in Section III.C.

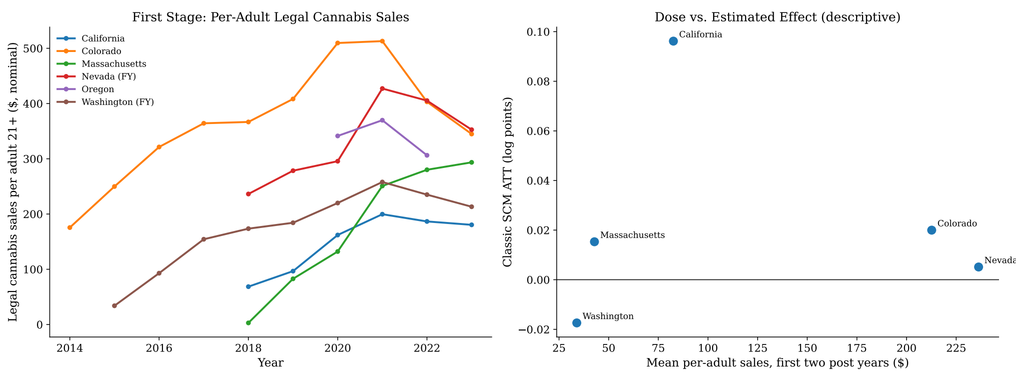

Figure 2. First Stage: Legal Cannabis Sales and Estimated Effects by Dose

Notes: Left: annual legal cannabis sales per adult aged 21 and older, nominal dollars, from state tax and regulatory portals; Washington and Nevada report fiscal years. California's figure is taxable sales and therefore overstates the change in effective access given pre-existing medical availability. Right: each state's classic synthetic control estimate against its mean per-adult sales in the first two retail years. The right panel is a five-point diagnostic; no slope is estimated from it.

C. Donor Pool and Covariates

The primary donor pool is the 30 states that had no recreational retail through 2023. I use never-treated states rather than not-yet-treated states because a donor that legalizes later in the sample enters the comparison partially treated, which biases the estimated effect toward zero. As a robustness check I also report a timing-adjusted pool that admits states legalizing after the 2000–2019 window

[1] Colorado Department of Revenue marijuana sales reports; Washington State Liquor and Cannabis Board sales files (fiscal years); Oregon Liquor and Cannabis Commission legislative reports (rounded annual totals only, as the agency's dashboard does not export data); Nevada Department of Taxation and Cannabis Compliance Board releases (fiscal years); California Department of Tax and Fee Administration quarterly taxable sales; Massachusetts Cannabis Control Commission open data. Sources, retrieval dates, and caveats for every figure are documented in the replication repository.

closes. The District of Columbia is excluded throughout.

The never-treated pool is not a completely clean control in the strict sense. Cannabis diffuses nationally through medical programs, informal markets, and cross-border purchasing (Hansen, Miller and Weber, 2020), so the donor states are partially exposed to the treatment, a violation of the stable unit treatment value assumption that applies to the whole design rather than to any single estimate. Its direction is known: under true substitution, exposed donors drink less, every constructed counterfactual sits too low, and every treated-state gap is pushed upward, against finding substitution. Its magnitude is not identified. Section VI bounds the practical size by dropping the most exposed donors, and the positive gaps that appear later in the paper, California's in particular, should be read with this bias in mind.

The primary specification matches on the pre-treatment outcome path only, with no outside covariates.

The reason is a result in Kaul et al. (2022): when a synthetic control matches on all pre-treatment outcome lags, covariates receive zero weight, so adding them creates the appearance of covariate adjustment without affecting the estimate. The primary specification is outcome-only, and a lag-subset specification (odd-year outcome lags plus four covariates measured as the average over the three years before each state's own treatment) is reported as a robustness check in Section VI. The covariates are real GDP per capita (Bureau of Economic Analysis), the unemployment rate (Bureau of Labor Statistics), the share of the population aged 20–34 (Census Bureau population estimates), and the beer excise tax (Federation of Tax Administrators), each measured pre-treatment for each state instead of a fixed baseline year.

II. Empirical Design

A. Pre-Specified Sample Rules

The estimates from synthetic controls are only credible where the pre-treatment fit is credible (Abadie, 2021), and fit-based sample decisions made after seeing the estimates lead to cherry-picking states. To prevent this I fix two inclusion rules before estimating treatment effects, using only pre-treatment information, and apply them uniformly.

The first rule is a fit-quality screen: for a treated state to enter the pooled sample its classic synthetic control pre-treatment root mean squared prediction error (RMSPE) has to be no more than twice the median RMSPE of in-space placebos, where each never-treated donor is fitted identically, as if treated at the same date. This benchmark asks if the method tracks the treated state as well as it tracks states where there is nothing unusual to track. The factor of two was fixed before estimation and allows for some sampling variation in the placebo fits. Table 2 reports the screen. Alaska and Nevada fail at 4.3 and 5.1 times the placebo median; the five remaining states pass. The primary sample is therefore

California, Colorado, Massachusetts, Oregon, Washington.[2]

The second rule concerns Washington, which privatized liquor sales in June 2012, two years before its cannabis retail opening. It is important that a level shift from privatization is not mistaken for a treatment effect. I run a Chow forecast test for a 2012 break in Washington's donor-demeaned log ethanol over 2000–2013; the test does not reject ($p = 0.17$ for total ethanol), so Washington stays in the sample. Section VI additionally reports Washington on a spirits-excluded outcome, where any privatization artifact would be concentrated.

Table 2—Pre-Specified Fit Screen

State	T_0	Pre-RMSPE	Placebo median	Ratio	Own-SD ratio	Included
Colorado	2014	0.0184	0.0126	1.46	0.86	Yes
Washington	2014	0.0149	0.0126	1.19	0.46	Yes
Oregon	2015	0.0150	0.0123	1.22	0.29	Yes
Alaska	2016	0.0521	0.0121	4.29	0.93	No
Nevada	2017	0.0602	0.0118	5.08	1.43	No
California	2018	0.0071	0.0129	0.55	0.35	Yes
Massachusetts	2018	0.0068	0.0129	0.53	0.43	Yes

Notes: Classic synthetic control fitted on the outcome path only, never-treated donors, 2000–2019 window, with treatment at each state's retail year T_0. The placebo median is the median pre-RMSPE across the 30 never-treated donors fitted identically at the same T_0. The inclusion rule, fixed before estimation, is Ratio ≤ 2. The own-SD ratio (pre-RMSPE over the standard deviation of the state's own pre-period log outcome) is reported as a diagnostic only.

B. Six Estimators

All six estimators are applied to the pooled effect of retail opening on log per-capita ethanol over 2000–2019, for the five clean-fit treated states against the never-treated donor pool, with each state's treatment activating during its own retail year.

The primary estimator is the partially-pooled staggered augmented synthetic control of Ben-Michael, Feller and Rothstein (2022). It estimates one synthetic control per treated state while shrinking the state-level imbalances toward a pooled target, and it aggregates across adoption cohorts in event time, so it is built exactly for this setting. The standard errors reported for this estimator are the software's default, a wild bootstrap over units. A jackknife over units is computed alongside it, and Section V evaluates both together with a design-calibrated randomization test.[3]

[2]An alternative normalization, RMSPE relative to the standard deviation of the state's own pre-period outcome, is shown in the table for transparency. It would also exclude Colorado, not because Colorado fits poorly but because Colorado's pre-period series is nearly flat, so the denominator is small.

[3]The default is what `augsynth` 0.2.0, installed from GitHub at commit `7a90ea4`, returns when `summary()` is called without specifying an inference type: a weighted wild bootstrap with 1,000 replicates. The jackknife is requested with `inf_type = "jackknife"`.

Along with the primary estimator I also report five alternatives, specifically chosen so the identifying assumptions differ. Synthetic difference-in-differences (Arkhangelsky et al., 2021) reweights both units and time periods. Unlike synthetic control, it allows a level shift between treated and synthetic units. The estimator assumes simultaneous adoption, so I estimate it separately by adoption cohort on balanced blocks and aggregate by treated post-period cell counts, with placebo-variance inference.[4] The generalized synthetic control (Xu, 2017) fits an interactive fixed effects model with the factor number chosen by cross-validation (selected $r = 1$) and parametric-bootstrap inference. Matrix completion (Athey et al., 2021) imputes the counterfactual by nuclear-norm-regularized completion of the untreated outcome matrix. The group-time estimator of Callaway and Sant'Anna (2021), with never-treated controls and multiplier-bootstrap inference, represents the modern difference-in-differences approach and avoids the negative-weighting problems of two-way fixed effects under staggered adoption (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœ uille, 2020). Plain two-way fixed effects with state-clustered standard errors completes the set as the conventional benchmark.

C. Inference

Each estimator has its own native inference, which is reported with replication counts in Table 3. Along with this I add three other pieces. First, a joint randomization-inference test of the pooled null: I draw 5 pseudo-treated states from the 30 never-treated donors 500 times and assign them the real staggered adoption dates at random. Then I re-estimate the primary estimator, and find the observed pooled estimate in the resulting null distribution. This is the paper's preferred test of the joint null as it utilizes the same panel, dates and estimator as the real analysis. Its two approximations, that the fit screen of Section II.A is not reapplied inside the draws and that the treated states are treated as exchangeable with the donors, are taken up in Section V. Second, for the state-by-state estimates I report prediction intervals from the method of Cattaneo, Feng and Titiunik (2021).[5] Third, the power simulation of Section V measures what any of these tests can and cannot identify.

III. Results

A. Pooled Estimates

Table 3 and Figure 3 report the pooled estimates. Six estimators across various identification approaches return pooled estimates between +0.7 and +2.8 percent.

[4]The cohort estimates share the donor pool, so treating them as independent in the aggregated standard error is an approximation. The conclusions are insensitive to inflating this standard error substantially.

[5]Permutation inference with 30 donors has a hard p-value floor of $1/31 \approx 0.032$: no treated state can achieve a smaller p-value, so multiple-comparison corrections that require thresholds below the floor cannot be satisfied by construction.

Every point estimate is positive, which is the wrong sign for substitution; every confidence interval contains zero, and the two-sided randomization-inference test of the joint null gives $p = 0.17$ against a null distribution with standard deviation 0.020 log points. The null is not specific to any one estimator's identifying assumptions.

TABLE 3—POOLED EFFECT OF RECREATIONAL RETAIL OPENING: SIX ESTIMATORS

Estimator	ATT	(SE)	Percent	95% CI	Inference
Partially-pooled ASCM (primary)	0.0276	(0.0419)	+2.8	[-0.055, +0.110]	Wild bootstrap (default)
Synthetic difference-in-differences	0.0181	(0.0153)	+1.8	[-0.012, +0.048]	Placebo (200 reps/cohort)
Generalized synthetic control (IFE)	0.0179	(0.0207)	+1.8	[-0.023, +0.058]	Param. bootstrap (500)
Matrix completion	0.0077	(0.0201)	+0.8	[-0.032, +0.047]	Bootstrap (500)
Callaway–Sant'Anna	0.0124	(0.0122)	+1.3	[-0.011, +0.036]	Multiplier bootstrap (999)
Two-way fixed effects	0.0071	(0.0229)	+0.7	[-0.038, +0.052]	Clustered by state

Notes: Outcome is log per-capita ethanol (21+), 2000–2019. Treated states are California, Colorado, Massachusetts, Oregon, Washington (retail years 2014, 2014, 2015, 2018, 2018); donors are the 30 never-treated states. ATT in log points; the percent column is $e^{\text{ATT}} - 1$. Synthetic difference-in-differences is estimated by adoption cohort and aggregated by treated post-period cells. Joint randomization-inference test of the pooled null using the primary estimator: $p = 0.17$ (500 draws). The primary row's standard error is the software default, a wild bootstrap over units (1,000 replicates). Under a jackknife over units the same estimate has a standard error of 0.0222 and a 95% CI of $[-0.016, +0.071]$, still a null.

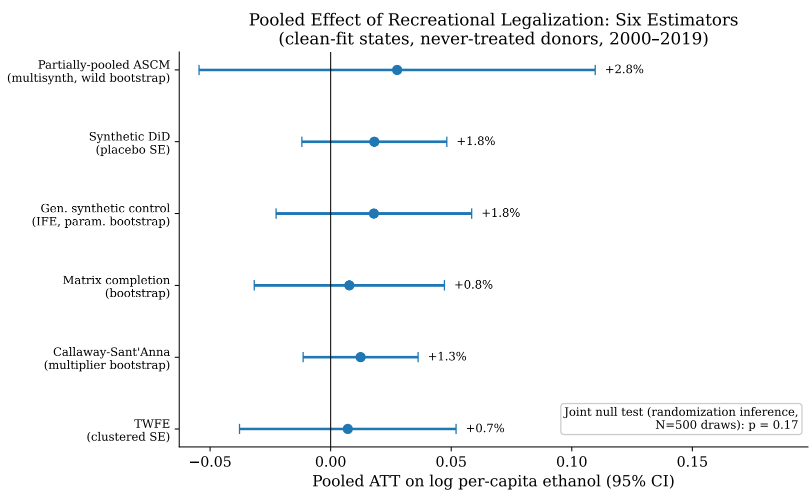

FIGURE 3. POOLED ESTIMATES: SIX ESTIMATORS

Notes: Pooled ATT estimates with 95% confidence intervals from Table 3. The annotation reports the randomization-inference p-value for the joint null under the primary estimator.

The Callaway–Sant'Anna event study (Figure 4) reaches the same conclusion.

Its post-treatment effects are scattered around zero with no trend. The pre-treatment coefficients are mostly flat, which is examined directly in Section IV.

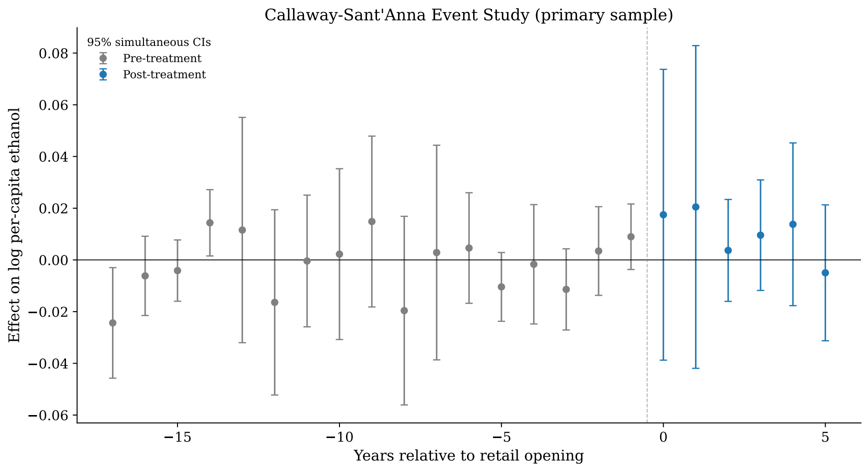

Figure 4. Callaway–Sant'Anna Event Study

Notes: Group-time average treatment effects aggregated to event time, never-treated control group, with 95% simultaneous confidence intervals from a multiplier bootstrap (999 iterations).

B. State-by-State Estimates

Table 4 reports the state-by-state synthetic control estimates, both regular and ridge-augmented, together with prediction intervals for post-period gaps. Colorado, the longest-exposed and highest-dose state, shows +2.0 percent with an interval which includes zero. Washington is the only state whose point estimate has the substitution sign (−1.7 percent), and its interval also covers zero. California is the outlier in the opposite direction and is discussed further in the next section.

C. California

California's gap is +10.1 percent, with alcohol sales above its synthetic control; Figure 5 shows that its prediction interval excludes zero in every post-treatment year, with a lower bound never below +5.2 percent, and Massachusetts shows the same pattern at a tenth the magnitude (+1.5 percent). Read causally, this would mean that legalization raised drinking in the two largest late-adopting states. Three features weigh against that reading. First, California's first-stage dose was the smallest in the sample (Section I.B), so the least-treated state carries the largest estimated effect, and the right panel of Figure 2 shows no dose gradient across states. Second, the sign matches the donor-contamination bias of

Table 4—State-by-State Synthetic Control Estimates

State	T_0	Pre-RMSPE	Classic	Ridge ASCM	Percent	Prediction interval
Colorado	2014	0.0184	0.0200	0.0200	+2.0	[-0.104, +0.099]
Washington	2014	0.0149	-0.0174	-0.0133	-1.7	[-0.152, +0.117]
Oregon	2015	0.0150	0.0217	0.0217	+2.2	[-0.099, +0.097]
Alaska	2016	0.0521	-0.0137	-0.0135	-1.4	—
Nevada	2017	0.0602	0.0051	0.0319	+0.5	—
California	2018	0.0071	0.0962	0.0962	+10.1	[+0.051, +0.131]
Massachusetts	2018	0.0068	0.0153	0.0153	+1.5	[+0.008, +0.024]

Notes: Classic and ridge-augmented synthetic control, outcome-only match, never-treated donors, 2000–2019. ATT in log points; the percent column applies to the classic estimate. The final column is the envelope of the 90% prediction intervals (Cattaneo, Feng and Titiunik, 2021) for the post-period gap across post years, for clean-fit states. Alaska and Nevada are shown for context only; they fail the fit screen and enter no pooled estimate.

Section I.C, which is largest for late adopters because their donors have had the longest exposure.[6] Third, the interval rests on two post-treatment years, 2018 and 2019. Regional drift, which produces negative pseudo-effects in Section IV, cannot explain a positive gap, so part of the gap remains unexplained single-unit noise. The same standard applies here as to the nulls: the estimate is reported, its known biases are stated, and it is not treated as a treatment effect.

The interval is not in conflict with the pooled minimum detectable effect of 6.4 percent from Section V; the two are different findings. The pooled figure is the power to detect a common effect across the five treated states under randomization inference. The interval is a single-unit statement about California's gap relative to its own synthetic, tight because California has the cleanest pre-treatment fit in the sample, and it does not absorb the regional drift that produces the one-to-two percent pseudo-effects of Section IV. The detectability benchmark for the paper's question is the pooled randomization power, not these single-unit intervals.

IV. In-Time Placebos

A constructed counterfactual is credible only if it does not find effects where no treatment occurred. Figure 1 shows that the gap between the treated states and the never-legalized average drifts over the course of the sample. If the estimator absorbs this drift imperfectly, it will find non-zero estimates in windows with no treatment. The size of these estimates is the noise floor any real estimator must be judged against. I measure this floor directly. First, I backdate the pooled design: all five clean-fit states are assigned a fake 2009 treatment on a panel that ends

[6]Contamination cannot account for the full gap, however. The synthetic control is a convex combination of donor outcomes, so a uniform downward contamination of donor log-alcohol by d raises the measured gap by about d. Reproducing California's roughly ten-log-point gap from contamination alone would require donor drinking to have fallen near ten percent relative to its uncontaminated path; since donors are only partially exposed to cannabis, that implies a full-legalization substitution effect well into double digits, larger than any estimate in the literature. Contamination plausibly explains the positive sign and a few points of the gap, not its full size.

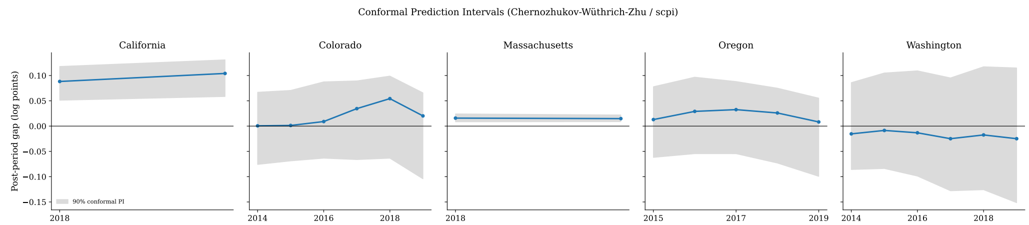

FIGURE 5. PREDICTION INTERVALS FOR THE POST-PERIOD GAPS

Notes: Post-period gaps (actual minus synthetic, log points) with 90% prediction intervals (Cattaneo, Feng and Titiunik, 2021), clean-fit states, Gaussian out-of-sample bounds, 200 simulations.

in 2013. This period sits entirely within every state's pre-treatment period, and every pooled estimator is re-run with its full inference. Second, for each clean-fit state, I estimate the fake effect at every feasible backdated adoption year (requiring at least six pre-treatment years and three fake post-treatment years, with data truncated before real treatment) for the classic, augmented, synthetic difference-in-differences, and generalized synthetic control estimators, thirty-nine fake estimates per method. This measures each estimator's noise in the panel.

TABLE 5—IN-TIME PLACEBOS: POOLED FAKE TREATMENT IN 2009

	Pooled fake $T_0 = 2009$			Single-state bias band	
Estimator	Fake ATT	(SE)	t	Mean	SD
Partially-pooled ASCM (primary)	-0.0116	(0.0226)	-0.51	-0.0085†	0.0281
Synthetic difference-in-differences	-0.0137	(0.0166)	-0.83	-0.0111	0.0229
Generalized synthetic control (IFE)	-0.0218	(0.0188)	-1.16	-0.0092	0.0340
Matrix completion	-0.0205	(0.0113)	-1.82	—	—
Callaway–Sant'Anna	-0.0090	(0.0122)	-0.74	—	—
Two-way fixed effects	-0.0205	(0.0137)	-1.49	—	—

Notes: Left panel: all five clean-fit states assigned a fake 2009 treatment, panel 2000–2013 (entirely pre-treatment), never-treated donors, full inference for each estimator. Right panel: mean and standard deviation of 39 single-state backdated placebo estimates per method (every feasible fake adoption year per state). The partially-pooled ASCM is a pooled estimator with no single-unit analogue; its cell (†) instead reports the classic single-treated-unit SCM band, the single-unit special case of the augmented family. Dashes mark estimators for which the single-state backdated exercise was not run (matrix completion, Callaway–Sant'Anna, two-way fixed effects). The partially-pooled ASCM standard error is the software-default wild bootstrap. Under a jackknife the pooled fake effects have $t = -0.77$ in the 2009 window and $t = -1.32$ in the 2007 window; neither rejects at the 5 percent level.

Table 5 and Figures 6–7 report the results. The latent-factor and time-weighted estimators do not absorb the regional drift that affects our convex-weight synthetic control. Every pooled estimator produces a negative fake effect on 2009–2013, ranging from -2.2 to -0.9 percent, with matrix completion closest to rejecting ($t = -1.82$). None of the six estimators reject at the five-percent level, and the pooled magnitudes are roughly half those of the single-state classic synthetic control placebos. Most critically, the pseudo-effects do not disappear under any estimator. The bias distributions show the same finding from the other direction: fake effects center near -0.9 percent with standard deviations of 2.3 to 3.4 percentage points. This pooled result is not unique to the 2009 window; if we repeat the exercise with a fake 2007–2011 adoption panel, it once again gives uniformly negative estimates, between -2.0 and -1.2 percent. In this window Callaway–Sant'Anna rejects the null at the 5 percent level ($t = -2.22$) on data where no treatment occurred. This is a false positive that anticipates the over-rejection its standard-error test shows in the power simulation of Section V. Any true effect smaller than two to three percent is inside the range these designs produce when no treatment occurred. This sets up the question the next section answers: given this noise floor, what effect sizes could the design detect?

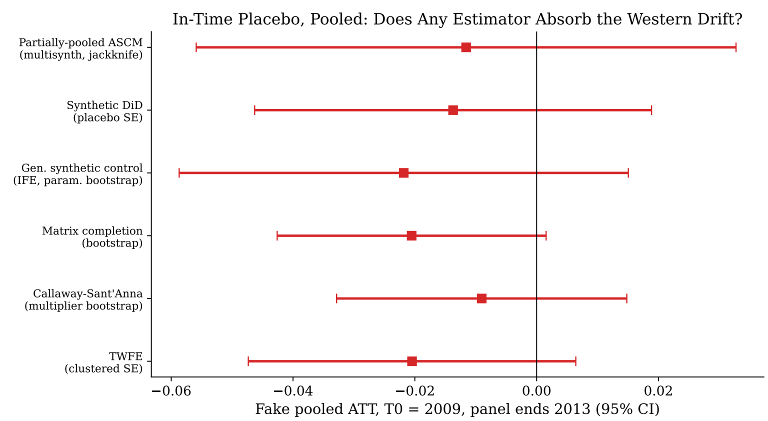

Figure 6. Pooled Fake-2009 Placebo Across Estimators

Notes: Fake pooled ATT with 95% confidence intervals, all clean-fit states assigned $T_0 = 2009$, panel ending 2013.

V. Statistical Power and the Minimum Detectable Effect

The placebo evidence gives the design's noise floor. This section asks the logical next question: if substitution of a given size were real, how often would this design detect it?

A. *Simulation Design*

The simulation preserves the design's panel, donor pool, staggered adoption dates, estimators, and inference rules. Each draw selects 5 pseudo-treated states at random from the 30 never-treated donors, assigns them the real staggered retail years (2014, 2014, 2015, 2018, 2018) in random order, and injects a multiplicative

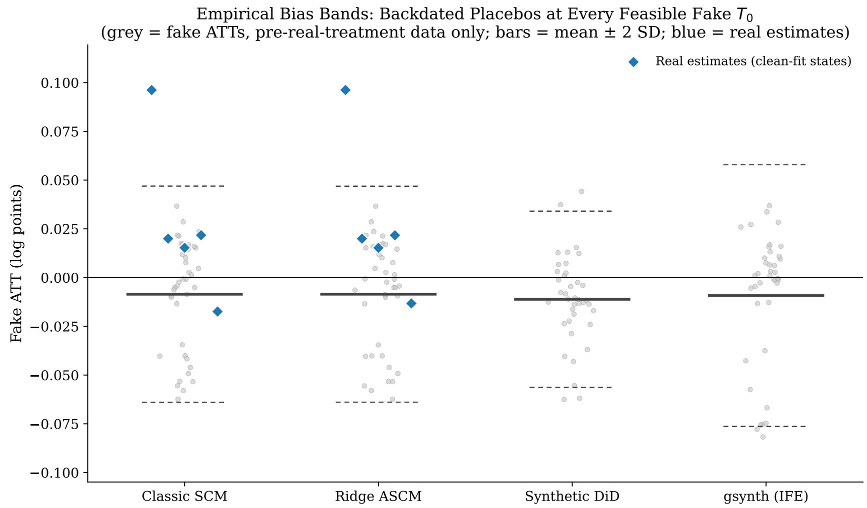

FIGURE 7. EMPIRICAL BIAS BANDS FROM BACKDATED PLACEBOS

Notes: Gray points are fake ATT estimates at every feasible backdated adoption year for every clean-fit state, using only pre-real-treatment data; bars mark the mean and ±2 SD of each estimator's fake distribution. Blue diamonds are the real estimates for the clean-fit states.

effect δ into their post-treatment outcomes ($\ln y$ += $\ln(1+\delta)$ for $t \geq T_0$). I draw pseudo-treated states from the never-treated pool, to guarantee that the outcomes contain no true treatment effect, so any rejection at $\delta = 0$ is a false positive and the rejection rate at $\delta < 0$ is power.[7] I re-run each estimator exactly as in Section III. The grid $\delta \in \{0, -2, -5, -8, -12\}$ percent spans the literature, from the low-single-digit effects that the survey and smaller quasi-experimental evidence suggest to the twelve percent that Baggio, Chong and Kwon (2020) estimate from scanner data. The primary estimator receives the grid with 400 draws per point and the four alternatives with 200, so every estimator gets a power curve and a minimum detectable effect (Bloom, 1995). A final cell repeats the -5 percent case with the effect phasing in linearly over three years, because a constant effect from the opening year is the most detectable shape and the first-stage evidence shows retail markets ramp up gradually.

Two features of the real procedure are not reproduced inside the draws. The first is the fit screen. The screen of Section II.A removed two of the seven treated states before pooling, but it is not reapplied to the pseudo-treated draws here

[7]The simulated power is an upper bound on the actual design's power, so the detectability conclusion is conservative. The pseudo-treated states are drawn from the never-treated donors, which are less drift-prone than the Western states that actually legalized (Figure 1), so the simulation likely understates the noise the real estimator faces. The donor draws also fit their synthetic controls better than the real treated states do, even after the fit screen of Section II.A keeps only the cleanest of those states, so the injected effect sits on top of less pre-existing prediction error. Both features push the reported power above what the real design achieves.

or in the randomization-inference draws of Section II.C. To address this I reran the simulation with the screen applied inside each draw, using the donor-placebo fit statistic of Section II.A. The screen binds often: 26 of the 90 donor and adoption-year cells fail it, and 86 percent of the draws lose at least one pseudo-treated state. This leads to power falling, because the surviving pooled samples are smaller than five states: the calibrated minimum detectable effect rises from 6.4 to 7.4 percent, the jackknife's from 7.2 to 8.1 percent, and the default bootstrap still rejects nothing at any effect. The screened randomization test gives $p = 0.21$ against the committed 0.17, both null. The five-state draws reported in the tables match the real analysis's post-screen count of five treated states, while the screened rerun applies the screen a second time inside the draw, so the two runs bracket the procedure; every conclusion in this section holds under the stricter one, and the reported curves are the more lenient reading. The second is exchangeability. The randomization test and the power draws treat the treated states as interchangeable with the donors, and Figure 1 shows they are not: the legalizing states are the drifting Western states, and pseudo-treated sets drawn from the donor pool face a better-behaved panel than the real treated set did. Restricting the pseudo-treated draws to Western donors is not available in this design, because the treated states absorb nearly the whole Census West and the never-treated pool retains only four Western states (Hawaii, Idaho, Utah, and Wyoming), too few to supply five pseudo-treated states. The in-time placebos of Section IV stand in for that check: they show what the estimators produce on the real treated states in windows with no treatment, a negative one-to-two percent floor. For the joint null test the direction is not problematic, since a null distribution shifted below zero and widened by drift makes the positive pooled estimate less extreme, so the reported failure to reject at $p = 0.17$ would survive; for power it is the direction footnote 7 records, and the reported curves are an upper bound.

For the primary estimator I evaluate three rejection rules. The first is a t-test on the software-default standard error, a wild bootstrap over units, exactly as the procedure runs when the inference type is left unspecified. The second is a t-test on a jackknife over treated units computed from the same fitted draw. The third is a design-calibrated randomization test that rejects when the absolute estimate exceeds the 95th percentile of the same estimator's $\delta = 0$ distribution. The alternative estimators are evaluated with their native standard-error tests and the calibrated rule. The calibrated rule has exact 5 percent size by construction, so its power column isolates the estimator's signal-to-noise from any miscalibration of its standard errors.

B. Results

Table 6 contains two main findings. First, the mean estimate tracks the injected effect almost exactly (-0.048 against a true -0.050; -0.080 against -0.080), so the estimator is approximately unbiased in this design. However, the test conven-

Table 6—Design-Based Power of the Primary Estimator

True δ	Draws	Reject at the 5% level: Default bootstrap	Jackknife	RI-calibrated	Mean estimate
0%	400	0.0%	1.8%	5.0%	0.0036
-2%	400	0.0%	7.8%	10.5%	-0.0166
-5%	400	0.0%	53.2%	63.7%	-0.0477
-8%	400	0.0%	89.2%	98.0%	-0.0797
-12%	400	0.0%	98.2%	100.0%	-0.1242
-5% (3-yr phase-in)	400	0.0%	23.2%	29.8%	-0.0319

Notes: Partially-pooled augmented synthetic control on pseudo-treated never-treated states with the real staggered dates. Rejection columns report the share of draws rejecting at the 5 percent level. The default-bootstrap and jackknife rules reject when $|\text{ATT/SE}| > 1.96$, using the software-default wild bootstrap standard error (as conventionally run) and the jackknife standard error, both computed on the same fitted draw. The RI-calibrated rule rejects when $|\text{ATT}|$ exceeds the 95th percentile of the estimator's own $\delta = 0$ draws (size 5% by construction). The jackknife size at $\delta = 0$ has a 95 percent binomial confidence interval of 0.7 to 3.6 percent. The mean-estimate column shows that the estimator recovers the injected effect nearly without bias. In the final row the -5 percent effect phases in linearly over three years ($\delta/3$, $2\delta/3$, then full) instead of arriving at once.

tionally obtained for this estimator, a t-test on the wild bootstrap standard error, rejects in 0.0 percent of draws at $\delta = 0$ and at every other value of δ. Zero rejections in 400 draws even when the true effect is -12 percent, three to four times the low-single-digit effects the survey evidence implies and at the highest end of anything the literature reports. This failure stems from the standard error rather than the point estimate; at $\delta = 0$ the mean default-bootstrap standard error is 1.7 times the true sampling dispersion of the estimator, and the inflation grows with the injected effect, reaching 6.5 times at $\delta = -12$ percent, so the t-statistic stays below 1.96 at every effect size. Section V.C traces this growth to the resampling scheme itself: the estimated effect enters the bootstrap's replicate variance, so the squared standard error rises one-for-one with the squared estimate and the t-statistic is bounded near one by construction (Appendix D derives this from the package's code path). The jackknife on the identical draws behaves differently: its mean standard error is a flat 1.27 times the true sampling dispersion at every δ, its empirical size at $\delta = 0$ is 1.8 percent, and its power column in Table 6 rises with the injected effect, giving a minimum detectable effect of 7.2 percent against the calibrated rule's 6.4, a gap that partly reflects its conservatism. With our five treated units, a null obtained from this estimator's default inference carries close to no evidential weight, here or in other applications of the method at this scale. Appendix Figure C1 compares the three rejection rules for the primary estimator directly.

The second finding is to do with the data itself. Under the correctly sized randomization rule, the primary estimator's power is 10 percent against a two percent effect, 64 percent against 5 percent, 98 percent against eight percent, and 100 percent against twelve percent (Figure 8). Interpolating, its minimum detectable effect at eighty percent power is 6.4 percent of per-capita ethanol.

At the clean-fit states' 2013 baseline of 2.84 gallons of pure ethanol per adult, that is roughly 3.2 standard drinks per adult per month. Table 7 reports the same grid for the alternatives. The standard-error tests of synthetic difference-in-differences, Callaway–Sant'Anna, and matrix completion over-reject at $\delta = 0$ (empirical sizes of 8, 10, and 12 percent against a nominal 5), so their apparent power partly reflects mis-calibration rather than signal. Under the calibrated thresholds the most sensitive estimator in the set is Callaway–Sant'Anna, with a minimum detectable effect of 4.5 percent. That figure is the design's limit across the estimators considered, and it is the relevant benchmark for what these methods can show on this data. Detection fails asymmetrically across the range of injected effects from the literature.[8] If we look back at the literature, an effect of twelve percent, the high end and the magnitude Baggio, Chong and Kwon (2020) report, would be caught by every calibrated test. An effect of two percent is far below every threshold: the most sensitive estimator rejects it in 26 percent of draws and the primary estimator in 10 percent. An effect of four percent is only detectable in the sharpest estimator, caught about 70 percent of the time, short of the 80 percent convention, while the primary estimator detects it in 46 percent. The low single digits, where the survey evidence and the smaller quasi-experimental estimates point, are therefore at or below what this design can see. The phase-in row of Table 6 adds a caution in the same direction: a -5 percent effect that ramps in over three years is detected in 30 percent of draws, against 64 percent for the constant version, so a realistic adoption path lowers power even further.

C. Why the Default Bootstrap Cannot Reject

The zero-power result has a mechanical explanation in the resampling method instead of the estimator or data. The default inference draws one mean-zero, unit-variance wild weight Z_i per state (a two-point Mammen weight, drawn for every unit in the panel, treated and donor alike) and computes each bootstrap replicate by reweighting unit-level outcome paths. Each treated unit's gap enters with weight Z_i, each donor's residual enters through the synthetic-control weights times Z_i, and the replicate is recentered by $(\sum_i Z_i/n_1)\,\widehat{\mathrm{ATT}}$, where n_1 is the number of treated units and the sum runs over all units. The weights multiply outcome levels instead of residuals centered at the estimate. An injected common effect cancels from the treated units' own term, because it moves each gap and the pooled estimate equally. It survives in the recentering term, which every unit

[8]The literature does not provide a single aggregate elasticity to compare against. The quasi-experimental estimates span null effects (Veligati et al., 2020; Armstrong et al., 2025), mixed beverage- and state-specific results (Calvert and Erickson, 2021), and the twelve percent retail decline of Baggio, Chong and Kwon (2020), with Anderson, Hansen and Rees (2013) finding substitution concentrated in beer and among young adults. Survey studies report that a large share of cannabis users reduce their drinking (Subbaraman, 2016), but user-level substitution rates need not aggregate to large population effects once cannabis-use prevalence and intensity are accounted for. I therefore treat the low single digits as the relevant small-effect case rather than as a point estimate from any one study.

Table 7—Power and Minimum Detectable Effects by Estimator

	Size, SE test	Power, RI-calibrated test				
Estimator	$\delta = 0$	-2%	-5%	-8%	-12%	MDE
Partially-pooled ASCM (primary)	0.0%	10.5%	63.7%	98.0%	100.0%	6.4%
Synthetic difference-in-differences	8.0%	17.5%	85.5%	100.0%	100.0%	4.8%
Generalized synthetic control (IFE)	3.5%	14.5%	71.0%	98.5%	99.5%	6.0%
Matrix completion	12.0%	12.5%	65.5%	96.0%	100.0%	6.4%
Callaway–Sant'Anna	10.0%	26.0%	92.0%	100.0%	100.0%	4.5%

Notes: Entries are the share of draws rejecting at the 5 percent level; 400 draws per cell for the primary estimator, 200 for the alternatives. The size column uses each estimator's native standard errors (with reduced replication counts inside the simulation: 25 placebo replications per cohort for synthetic difference-in-differences, 100 bootstrap draws for the generalized synthetic control and matrix completion, 199 multiplier iterations for Callaway–Sant'Anna). For the primary estimator the native standard error is the software default, a wild bootstrap over units; Table 6 reports the jackknife for that estimator. The power columns use each estimator's own $\delta = 0$ 95th percentile as the rejection threshold, which has 5% size by construction. The MDE column interpolates the calibrated power curve to 80 percent power.

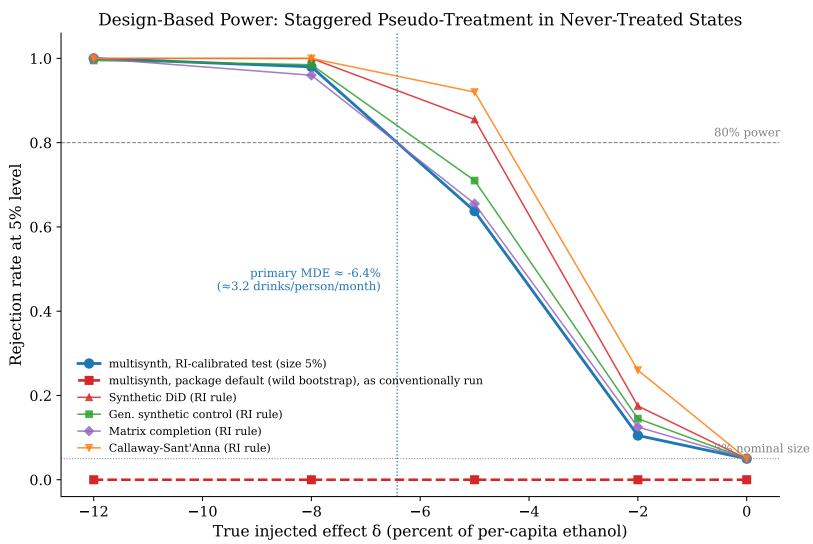

Figure 8. Power Curves and the Minimum Detectable Effect

Notes: Rejection rates at the 5% level against the injected effect δ. The heavy solid curve is the primary estimator under the design-calibrated randomization test; the dashed curve is the same estimator under the software default (a wild bootstrap), as conventionally run, which never leaves zero. The thin curves are the four alternative estimators under their own calibrated thresholds. The dotted vertical line marks the primary estimator's minimum detectable effect at 80 percent power.

carries: each of the n_0 donor states contributes $Z_d\,\widehat{\text{ATT}}/n_1$ to the replicate, and those contributions do not cancel in the variance. The replicate is exactly linear in the weights, so its variance can be written out in full:

$$\text{Var}^*\big(\widehat{\text{ATT}}^*\big) \;=\; \tilde{\sigma}^2 \;+\; \frac{2\,\widehat{\text{ATT}}}{n_1}\,C \;+\; \frac{n_0}{n_1^2}\,\widehat{\text{ATT}}^2,$$

where $\tilde{\sigma}^2$ collects the effect-free fit-noise terms and C is a fit-noise cross-term; Appendix D derives this line by line from the package source. The quadratic coefficient is n_0/n_1^2. Inside the simulation the five pseudo-treated states are drawn out of the thirty-donor pool, so $n_0 = 25$, $n_1 = 5$, and the coefficient is exactly one; for the real-data estimate $n_0 = 30$ and the coefficient is 1.2.

The simulation draws show this relationship clearly. Regressing the squared bootstrap standard error on the squared estimate across all 2,000 primary-estimator draws of the power simulation gives a slope of 1.01 against the derived coefficient of one, with a correlation of 0.99; the jackknife standard error computed on the identical fitted draws shows no such relationship (correlation 0.05). The bounded t-statistic follows: the test statistic is $|\widehat{\text{ATT}}|/(\tilde{\sigma}^2 + n_0\widehat{\text{ATT}}^2/n_1^2)^{1/2}$ up to the cross-term, which increases toward an asymptote of $n_1/\sqrt{n_0} = 1$ as the effect grows. The largest t-statistic the default test produced across the 2,000 draws is 1.08, and its average at the largest injected effect, $\delta = -12$ percent, is 0.97. The critical value of 1.96 is unreachable at any effect size, so the size of the default test at $\delta = 0$ is exactly zero, as opposed to some small number, and its power is zero against every alternative by construction. This is a stronger statement than under-rejection: the test cannot reject, no matter how large the true effect.

The problem originates from the resampling scheme, not its execution and not this specific paper's data, method or topic. The package computes exactly what the scheme prescribes, applying mean-zero weights to outcome levels and recentering by $(\sum_i Z_i/n_1)\,\widehat{\text{ATT}}$, and the calls here follow the documented default. The behavior was verified against the package source at the commit recorded in Section II.B, the replication materials reproduce it at the level of the paired draws, and no released or development version of the package changes this default as of July 2026. The scheme is unusable as a t-test standard error when treated units are few, because the recentering plants the estimate in every replicate. A wild bootstrap on residuals centered at the estimate would not have the property, and the jackknife, which is invariant to a common shift in the treated units' post-period gaps, is immune on the same fitted draws, which is why its standard error is flat across the grid. This finding is an extreme version of a known one: MacKinnon and Webb (2018) show that wild cluster bootstrap tests can under-reject badly when few clusters are treated because the treated clusters dominate the resampled statistic; here the effect enters the replicate variance directly, and under-rejection turns into an inability to reject anything.

VI. Robustness

Table 8 re-estimates the primary specification varying one ingredient at a time. The pooled null is stable across all variations.

TABLE 8—ROBUSTNESS OF THE POOLED ESTIMATE

Specification	ATT	(SE)	Percent
Primary specification	0.0276	(0.0419)	+2.8
Drop donors adjacent to treated states	0.0289	(0.0397)	+2.9
Drop Oklahoma (high-intensity medical)	0.0276	(0.0411)	+2.8
Drop New Hampshire (cross-border sales)	0.0276	(0.0431)	+2.8
Timing-adjusted donor pool	0.0254	(0.0464)	+2.6
Extension window, 2000–2023	0.0416	(0.0627)	+4.2
All seven treated states	0.0225	(0.0318)	+2.3
Beer ethanol only	0.0385	(0.0521)	+3.9
Wine ethanol only	-0.0295	(0.0691)	-2.9
Spirits ethanol only	0.0340	(0.0490)	+3.5
Washington, spirits-excluded outcome	-0.0289	[-0.100, +0.038]	-2.9

Notes: All rows except the last are the partially-pooled augmented synthetic control with the package-default wild bootstrap standard errors, varying one ingredient at a time from the primary specification. Every variant remains a null under the jackknife as well; those standard errors are in the replication materials (`robustness_jack.csv`). The Washington spirits-excluded row is a single-state ridge-augmented estimate on beer-plus-wine ethanol with a jackknife+ 95% interval in place of a standard error.

Spillovers and contamination. Donor states that are adjacent to any treated state are exposed to cross-border cannabis purchasing (Hansen, Miller and Weber, 2020), and high-intensity medical-cannabis states are partially treated in substance. Dropping the adjacent donor states, and separately removing Oklahoma, whose post-2018 medical program produced the country's highest dispensary density, moves the pooled estimate by less than a tenth of a standard error. So does dropping New Hampshire, the donor whose state-run, untaxed liquor sales attract the heaviest cross-border purchasing and which carries large weights in several synthetic controls (Appendix Table A1). The residual bias runs in the direction stated in Section I.C, toward positive gaps and against finding substitution, and is the likely source of the uniformly positive point estimates.

Donor pool and window. The timing-adjusted donor pool (admitting states that legalized after 2019) and the full 2000–2023 extension window (which adds the COVID years, +4.2 percent) leave the conclusion unchanged, as does re-admitting Alaska and Nevada in defiance of the fit screen.

Beverage decomposition. If we look more granularly at the types of alcohol by pooling estimates by beverage, we find that the attached default wild bootstrap standard errors give positive nulls for beer and spirits and a negative estimate for wine (−2.9 percent). Wine is the only beverage whose sign matches substitution, and its magnitude sits inside the range the power analysis shows this design cannot distinguish from zero. This is a demonstration of the paper's whole point, just at a smaller scale: a substitution-signed estimate of plausible size is exactly what an underpowered design produces by chance.

Washington and privatization. On the spirits-excluded outcome, Washington's estimate is -2.9 percent with an interval of $[-0.100, +0.038]$ log points. This sign matches substitution, but the interval contains zero comfortably.

Covariates. The lag-subset specification (odd-year lags plus four pre-treatment covariates, each predictor standardized across donors) yields state-level estimates more spread apart than the outcome-only match but with the same pooled message (Appendix Table B1). The added dispersion is consistent with Kaul et al. (2022): covariate matching changes the appearance of the specification without improving it.

VII. Discussion

There are three main results which structure the extent of what this paper can say. The pooled estimates show that the aggregate null holds across identification approaches, so estimator choice does not explain the null in this design. The placebo analysis shows that every one of those estimators produces one-to-two percent pseudo-effects on no-treatment data in this setting, so estimates of that magnitude, at the small end of the low single digits where plausible substitution would lie, are inside the design's noise floor. The power analysis makes the same point: even with correctly calibrated inference, no estimator in the set can detect effects smaller than 4.5 percent at conventional power, and the software-default inference attached to the primary estimator detects nothing at all.

For the existing literature the result is a caution, not a verdict. The null estimated here is evidence about detectability, not evidence that cannabis–alcohol substitution does not occur. The granular studies that find substitution (Baggio, Chong and Kwon, 2020; Anderson, Hansen and Rees, 2013), the mixed household-purchase results of Calvert and Erickson (2021), and the aggregate studies that find nothing (Veligati et al., 2020; Armstrong et al., 2025) are consistent with each other: a 12 percent scanner-data effect concentrated in specific products and places is compatible with an aggregate effect in the low single digits that no state-panel design of this size can see. Other aggregate studies differ in data, treatment definitions, estimators, sample periods, and donor pools, and their power is their own question; this paper's simulation does not answer it for them, though the configuration audited here is common enough that the question is valuable. The policy implication is correspondingly bounded: on the evidence a design of this power can produce, claims that legalization measurably reduces drinking, and claims that it measurably does not, both outrun the data.

The methodological contribution is more applicable. Staggered synthetic-control designs with a handful of treated units are now common, and their null results are routinely reported with jackknife or placebo-based uncertainty. The simulation here shows why the audit is necessary: in one standard configuration of that machinery (five treated units, thirty donors, twenty annual periods), the estimator is nearly unbiased while its default inference is structurally incapable of rejecting, and nothing short of a power audit reveals this. The few-treated-clusters literature (Conley and Taber, 2011; Ferman and Pinto, 2019; MacKinnon and Webb, 2018) documents size distortions similar to this type; the audit is what turns those warnings into a statement about a specific design, and here it found a failure far stronger than under-rejection. The jackknife and calibrated randomization inference both have usable size and power on the same design, but neither can see low-single-digit effects, so the detectability conclusion does not depend on the choice of tests. A design-based power statement of the kind in Section V is cheap to produce and should accompany null results when this research design is used.

The limitations are the familiar ones from the cannabis–alcohol substitution literature. The outcome measures sales, not consumption, and attributes cross-border purchases to the selling state. The treatment is binary retail opening, while effective dose varies by an order of magnitude across states; the first-stage section documents this but the pooled estimators do not exploit it, and a dose-response design is a natural next step. It is possible that the donor pool is contaminated by national cannabis diffusion, with a known anti-substitution bias direction but unknown magnitude. The power simulation inserts multiplicative effects that are constant or follow a simple three-year phase-in; richer dynamics would change the numbers, but it is worth noting that the phase-in result suggests that added realism lowers power rather than raising it. This leaves the conclusion that effects in the low single digits are out of reach factually intact.

VIII. Conclusion

This paper estimates the effect of recreational cannabis retail openings on state-level alcohol sales and asks what such a design can actually detect. The findings are consistent across three exercises. Six estimators return a range of small, positive-signed nulls. Backdated placebos show that every estimator produces spurious effects of the same size as the substitution effects from the academic literature. The power simulation finds that the primary estimator's software-default inference rejects nothing at any effect size, while even correctly calibrated inference cannot detect effects below 4.5 percent of per-capita ethanol.

The null found here is therefore not evidence against substitution, and aggregate designs of similar power face the same limit, which a simulation of this kind can check. What twenty years of state-level sales data can establish is real: it rules out large aggregate effects, including the twelve-percent-scale decline found in scanner data, but it cannot resolve the low-single-digit effects that plausible substitution would produce.

The findings from this paper extend past cannabis. Only a power statement reveals whether a null from this design carries any information: run with its software-default inference, a small-sample staggered synthetic control was close to guaranteed to find nothing here, and even the tests that work cannot see effects in the low single digits. Detecting effects of plausible size will require what the studies that find substitution already use: granular outcomes, credible dose variation, and designs whose power is demonstrated.

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Donor Weights and Covariate Balance

Table A1—Synthetic Control Donor Weights

Treated state	Donors (weight)
Alaska	Texas (0.661), New Hampshire (0.328), Kansas (0.011)
California	Florida (0.271), Minnesota (0.195), Georgia (0.133), Utah (0.116), South Carolina (0.109), Louisiana (0.102), North Dakota (0.032), Tennessee (0.029)
Colorado	Georgia (0.346), New Hampshire (0.232), Minnesota (0.173), Wisconsin (0.102), Louisiana (0.072), Florida (0.063), North Carolina (0.012)
Massachusetts	South Carolina (0.192), Texas (0.150), New Hampshire (0.135), Georgia (0.131), Florida (0.126), Hawaii (0.090), Tennessee (0.081), Louisiana (0.054)
Nevada	New Hampshire (0.668), Georgia (0.332)
Oregon	Hawaii (0.278), Louisiana (0.234), North Dakota (0.180), Kentucky (0.100), Iowa (0.075), Florida (0.052), Delaware (0.048), Wyoming (0.033)
Washington	South Carolina (0.390), Hawaii (0.325), Kansas (0.266), Minnesota (0.019)

Notes: Classic synthetic control weights (outcome-only match, never-treated donors, 2000–2019), donors with weight above 0.001, largest eight per state.

Table A2—Covariate Balance, Pre-Treatment

State (T_0)	Covariate	Treated	Synthetic	Donor average
Colorado (2014)	GDP per capita ($)	56,117	52,998	51,824
	Unemployment (%)	7.8	7.2	6.8
	Share aged 20–34	0.214	0.198	0.202
	Beer tax ($/gal)	0.08	0.27	0.29
Washington (2014)	GDP per capita ($)	61,728	49,513	51,824
	Unemployment (%)	8.2	7.0	6.8
	Share aged 20–34	0.209	0.199	0.202
	Beer tax ($/gal)	0.76	0.65	0.29
Oregon (2015)	GDP per capita ($)	48,579	58,876	52,601
	Unemployment (%)	7.7	5.3	6.1
	Share aged 20–34	0.205	0.209	0.203
	Beer tax ($/gal)	0.08	0.42	0.30
Alaska (2016)	GDP per capita ($)	73,024	58,017	53,391
	Unemployment (%)	6.6	5.0	5.4
	Share aged 20–34	0.224	0.207	0.203
	Beer tax ($/gal)	1.07	0.23	0.31
Nevada (2017)	GDP per capita ($)	52,547	56,994	54,017
	Unemployment (%)	7.0	4.4	4.8
	Share aged 20–34	0.207	0.193	0.203
	Beer tax ($/gal)	0.16	0.31	0.32
California (2018)	GDP per capita ($)	67,376	53,414	54,436
	Unemployment (%)	5.5	4.6	4.3
	Share aged 20–34	0.221	0.205	0.202
	Beer tax ($/gal)	0.20	0.41	0.33
Massachusetts (2018)	GDP per capita ($)	76,441	53,523	54,436
	Unemployment (%)	4.2	4.5	4.3
	Share aged 20–34	0.213	0.201	0.202
	Beer tax ($/gal)	0.11	0.53	0.33

Notes: Covariates measured as each state's average over the three years before its own treatment year. Synthetic values use the classic outcome-only weights of Table A1; balance is therefore diagnostic (covariates are not matched on in the primary specification, per Section II).

Additional Robustness Tables

Table B1—Lag-Subset Specification with Pre-Treatment Covariates

State	T_0	Outcome-only ATT	Lag-subset + covariates ATT
Colorado	2014	0.0200	0.0212
Washington	2014	-0.0174	-0.0697
Oregon	2015	0.0217	0.0629
California	2018	0.0962	0.1057
Massachusetts	2018	0.0153	-0.0335

Notes: Classic synthetic control per clean-fit state. The outcome-only column matches on all pre-treatment lags; the lag-subset column matches on odd-year lags plus real GDP per capita, unemployment, the share aged 20–34, and the beer excise tax, each measured as the average over the three years before the state's treatment and standardized across donors (Kaul et al., 2022). ATT in log points.

Table B2—First-Stage Dose by State

State	Retail year	Sales per adult, first two years ($)
Nevada	2017	236
Colorado	2014	213
California	2018	83
Massachusetts	2018	43
Washington	2014	34

Notes: Mean annual legal cannabis sales per adult aged 21 and older over each state's first two retail years, nominal dollars, from state tax and regulatory sources. Oregon is omitted: its agency publishes only rounded totals beginning in 2020, after its retail opening. Washington and Nevada report fiscal years. California's figure is taxable sales.

Inference Rules for the Primary Estimator

Figure C1 compares the three rejection rules for the primary estimator on the same simulation draws. The software-default wild bootstrap never rejects on the grid. The jackknife tracks the calibrated randomization test from below, with its mild conservatism visible as the gap between the two curves and in its 7.2 percent minimum detectable effect against the calibrated rule's 6.4 percent.

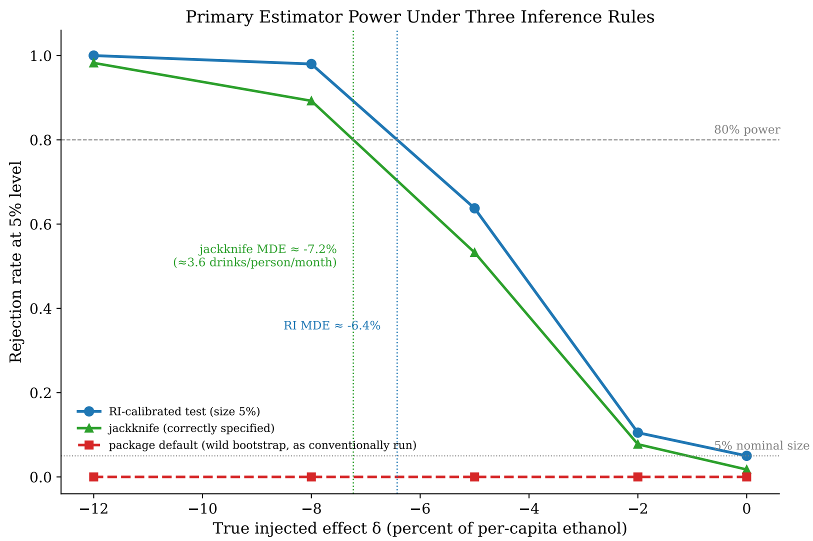

Figure C1. Three Rejection Rules for the Primary Estimator

Notes: Rejection rates at the 5% level against the injected effect δ, partially-pooled ASCM only, 400 draws per cell. The dotted vertical lines mark the minimum detectable effects at 80 percent power under the calibrated rule and the jackknife. Rejection rates by δ are in Table 6.

This appendix derives the replicate variance of the default inference exactly, tied to the code path of `augsynth` 0.2.0 at the commit recorded in Section II.B. When `summary()` is called on a `multisynth` fit without an inference type, the package calls `weighted_bootstrap_multi`, which draws $B = 1{,}000$ weight vectors and computes, for each, the statistic

$$\widehat{\mathrm{ATT}}^{*}(Z) \;=\; \mathrm{ATT}^{*}(Z) \;-\; \frac{\sum_{i=1}^{n} Z_i}{n_1}\,\widehat{\mathrm{ATT}},$$

where $Z = (Z_1, \ldots, Z_n)$ assigns one weight to every unit in the panel, treated and donor alike, drawn independently from the two-point Mammen distribution with mean zero and variance one, n_1 is the number of treated units, and $\mathrm{ATT}^*(Z)$ is the pooled estimate recomputed by `predict` with `bs_weight` $= Z$. The reported standard error is the standard deviation of $\widehat{\mathrm{ATT}}^{*}$ across the B replicates.

With staggered treated units each treated unit j forms its own cohort, and `predict` computes the replicate cohort effect as

$$\tau_j^*(Z) \;=\; Z_j\, g_j \;-\; \sum_d w_{dj}\, Z_d\, \bar{r}_{dj}, \qquad g_j = \bar{y}_j - \hat{y}_j^0,$$

where $\bar{y}_j$ is unit j's post-period mean outcome, $\hat{y}_j^0$ is its level-one prediction, w_{dj} are the synthetic-control weights, and $\bar{r}_{dj}$ is donor d's mean residual over unit j's post window. The pooled replicate is $\mathrm{ATT}^*(Z) = n_1^{-1}\sum_j \tau_j^*(Z)$. The weights multiply outcome levels ($Z_j g_j$), not residuals centered at the estimate.

Collecting terms, $\widehat{\mathrm{ATT}}^{*}(Z)$ is exactly linear in Z: treated unit j carries coefficient $(g_j - \widehat{\mathrm{ATT}})/n_1$, and donor d carries coefficient $-(c_d + \widehat{\mathrm{ATT}}/n_1)$ with $c_d = n_1^{-1}\sum_j w_{dj}\bar{r}_{dj}$. Because the Z_i are independent with unit variance, the replicate variance is the sum of squared coefficients:

$$\mathrm{Var}^*\big(\widehat{\mathrm{ATT}}^{*}\big) \;=\; \underbrace{\frac{1}{n_1^2}\sum_j \big(g_j - \widehat{\mathrm{ATT}}\big)^2 \;+\; \sum_d c_d^2}_{\tilde{\sigma}^2} \;+\; \frac{2\,\widehat{\mathrm{ATT}}}{n_1}\underbrace{\sum_d c_d}_{C} \;+\; \frac{n_0}{n_1^2}\,\widehat{\mathrm{ATT}}^{2}.$$

Two facts follow. First, an injected common effect τ cancels from the treated term: it raises every g_j and $\widehat{\mathrm{ATT}}$ by the same amount, so $g_j - \widehat{\mathrm{ATT}}$ is unchanged. A calculation that stops at the treated average, $\mathrm{Var}^* = n_1^{-2}\sum_j(\tau + e_j)^2 \approx \tau^2/n_1$, misses both the cancellation and the recentering term, and returns the wrong coefficient. Second, the effect enters through the recentering: every one of the n_0 donors carries $\widehat{\mathrm{ATT}}/n_1$, and the n_0 squared contributions sum to $(n_0/n_1^2)\,\widehat{\mathrm{ATT}}^{2}$.

In the power simulation the five pseudo-treated states are drawn out of the thirty never-treated donors, so each draw's panel has $n_1 = 5$ and $n_0 = 25$: the quadratic coefficient is $25/25 = 1$, and the t-statistic rises to an asymptote of $n_1/\sqrt{n_0} = 1$, up to the cross-term. In the real-data fit $n_0 = 30$, the coefficient is $30/25 = 1.2$, and the asymptote is $5/\sqrt{30} \approx 0.91$. Fitting the derived form to the 2,000 committed simulation draws gives

$$\widehat{\text{se}}^2_{\text{boot}} \;=\; 0.00084 \;+\; 0.98 \cdot \widehat{\text{ATT}}^2 \;+\; (-0.0034) \cdot \widehat{\text{ATT}}, \qquad R^2 = 0.98,$$

a quadratic coefficient of 0.98 against the derived value of one, with a small cross-term. The largest t-statistic across all 2,000 draws is 1.08; the excess over the asymptote reflects the cross-term and draw-level variation in $\tilde{\sigma}^2$. At the 5 percent critical value of 1.96 the rejection region is empty at every effect size: the size of the default test is exactly zero, and so is its power.